

Experiment 3

Natural Response (MATLAB)

1. Implement the following code in MATLAB

```
% NATURAL RESPONSE OF RC CIRCUIT
% Define the network parameters
C = 0.22e-6;
R = 47e3;

% Calculate the time constant
tau = R*C;

% Specify the initial voltage across the capacitor
V0 = 2;

% Define the points at which we will compute the current
t = 0 : 0.001 : 0.1;

% Calculate the current
I = V0/R*exp(-t/tau);

% Plot the responses
plot(t, I);
grid on
title('Natural response - RC circuit')
xlabel('Time (s)')
ylabel('Current (A)')
```

2. Modify the above code to also plot the voltage across the resistance.
3. Write a for loop to compute the current across 3 values of resistances: 1k Ω , 10k Ω , and 47k Ω .
4. Plot them in the same graph.
5. Could you repeat point 3 above without using a for loop and without having a separate section for each resistance? (*Hint*. Search for dot multiplication in MATLAB).